

TrES-1b

R-2433

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_100724 · created 2026-08-22T00:57:19+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455401.789 +/- 0.0006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1 +/- 0.038
Transit depth (Rp/Rs)^2	0.99 +/- 0.76 [%]
Orbital Inclination (inc)	85.0 +/- 1.53
Airmass coefficient 1 (a1)	0.3832 +/- 0.0056
Airmass coefficient 2 (a2)	0.91 +/- 0.29
Scatter in the residuals of the lightcurve fit is	1.46 %
Best Comparison Star	#2 - [408, 369]
Optimal Method	PSF photometry
Transit Duration (day)	0.027 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1 ± 0.038 vs 0.1358 (deviation 0.94σ)
Duration fit vs published	0.027 d vs 0.1042 d (deviation 4.06σ)
Mid-time O–C	0.6 min
Depth SNR	2.6
Residual scatter	1.46 %

Light curve

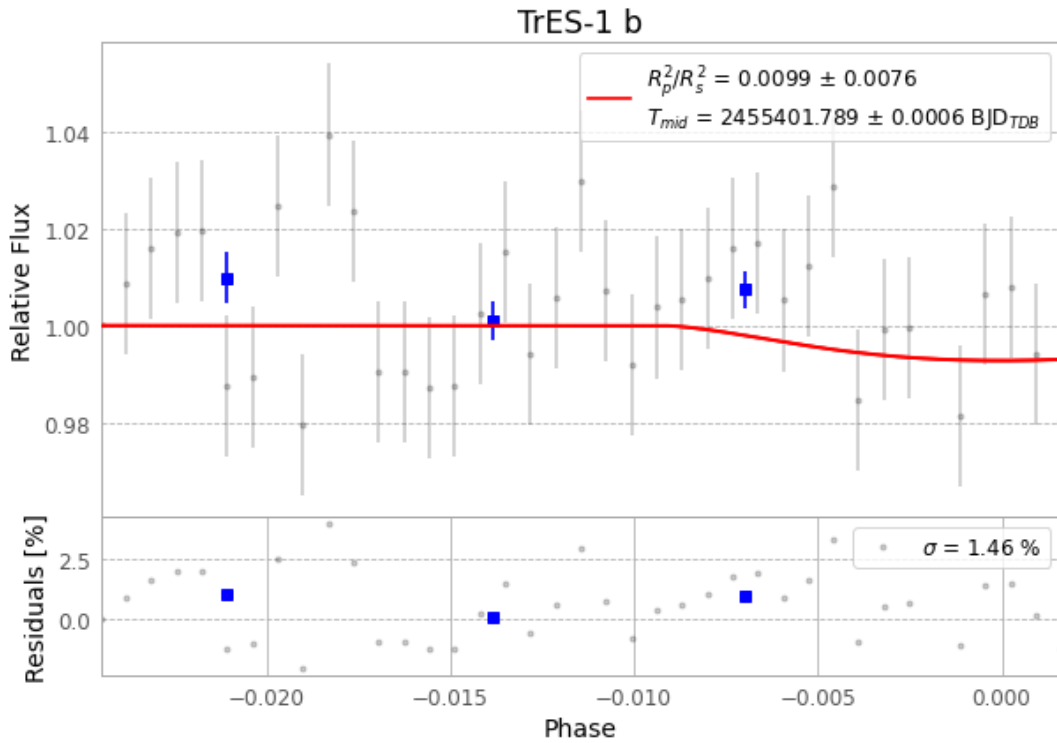


Figure 1 — FinalLightCurve_TrES-1 b_2010-07-23.png (SHA-256 0d5e26478cf425b4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [234, 191], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5d201097705c5ae6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

39 FITS frames (26 MB), TRES-1100724050315.FITS → TRES-1100724065714.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-100724.log (0729b98e66a8...), FinalLightCurve_TrES-1 b_2010-07-23.png (0d5e26478cf4...), FinalLightCurve_TrES-1 b_2010-07-23.csv (249357b48ed4...)

Compute resources

Wall time 76.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 00:57Z.